

Simone Alberto Peirone

ELLIS PHD STUDENT · ARTIFICIAL INTELLIGENCE

Turin, Italy

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About me

I am a fourth-year PhD Student in Artificial Intelligence at Polytechnic University of Turin (Italy), under the supervision of **Prof. Giuseppe Averta**. I am mostly interested in building **egocentric video understanding** models that mimic the compositionality and hierarchy behind human reasoning skills. In Summer 2025, I visited EPFL under the supervision of **Prof. Pascal Frossard**, exploring the intersection of video understanding and **concepts learning**.

Education

Polytechnic University of Turin

ELLIS PHD STUDENT

Turin, Italy

November 2022 - present

- Working on egocentric video understanding of human activities through the lenses of *compositionality*, exploring how different reasoning tasks can share and benefit from complementary knowledge, and *hierarchy*, by enabling models to reason about the structure behind human activities.
- **Keywords:** video understanding; concepts learning; graph neural networks.
- **Programme:** National PhD Programme in Artificial Intelligence
- **Thesis title:** Egocentric Vision for Advanced Human-Robot Cooperation
- **Supervisors:** Prof. Giuseppe Averta, Prof. Barbara Caputo

École Polytechnique Fédérale de Lausanne (EPFL)

VISITING PHD STUDENT

Lausanne, Switzerland

March 2025 - Sept. 2025

- Working on applications of concepts learning to video understanding models.
- **Supervisor:** Prof. Pascal Frossard

Polytechnic University of Turin

M.SC. COMPUTER ENGINEERING

Turin, Italy

2020 - 2022

- **Thesis title:** EGO-T³: Test Time Training for Egocentric Videos
- **Supervisors:** Prof. Barbara Caputo, Doct. Mirco Planamente, Doct. Chiara Plizzari
- **Final mark:** 110/110 cum Laude

Polytechnic University of Turin

B.SC. COMPUTER ENGINEERING

Turin, Italy

2017 - 2020

- **Final mark:** 110/110 cum Laude

Publications

Peirone, S. A., Senouf, O., Pistilli, F., Averta, G., Frossard, P. (2026). A General Concept-based Decomposition for Vision-Language Embeddings. **Annual Conference on Neural Information Processing Systems (NeurIPS)**.

Pistilli, F., **Peirone, S. A.**, Averta, G. (2026). Learning to Evolve Scenes: Reasoning about Human Activities with Scene Graphs **Annual Conference on Neural Information Processing Systems (NeurIPS)**. [Page] [Paper]

Peirone, S. A., Pistilli, F., Alliegro, A., Tommasi, T., & Averta, G. (2025). Hier-EgoPack: Hierarchical Egocentric Video Understanding with Diverse Task Perspectives. **IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)**. [Page] [Paper] [Code]

Peirone, S. A., Pistilli, F., & Averta, G. (2025). HiERO: understanding the hierarchy of human behavior enhances reasoning on egocentric videos. **IEEE/CVF International Conference on Computer Vision (ICCV)**. [Page] [Paper] [Code]

Nasirimajd, A., Plizzari, C., **Peirone, S. A.**, Ciccone, M., Averta, G. & Caputo, B. (2025). Domain Generalization using Action Sequences for Egocentric Action Recognition. **Pattern Recognition Letters (PRL)**. [Page] [Paper] [Code]

Peirone, S. A.*, Goletto, G.*, Planamente, M., Bottino, A., Caputo, B., & Averta, G. (2025). Egocentric zone-aware action recognition across environments. **Pattern Recognition Letters (PRL)**. [Page] [Paper] [Code]

Peirone, S. A., Pistilli, F., Alliegro, A., & Averta, G. (2024). A backpack full of skills: Egocentric video understanding with diverse task perspectives. **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)**. [Page] [Paper] [Code]

Planamente, M, Plizzari, C., **Peirone, S. A.**, Caputo, B., & Bottino, A. (2024). Relative norm alignment for tackling domain shift in deep multi-modal classification. **International Journal of Computer Vision (IJCV)**. [Paper]

Technical Reports

Zenotto, A., **Peirone, S. A.**, Pistilli, F., & Averta, G. (2026). HiERO-StepG@ Ego4D Step Grounding Challenge: hierarchical activity understanding enables zero-shot step grounding. **arXiv:2605.31227**. [Report]

Nasirimajd, A., **Peirone, S. A.**, Plizzari, C., & Caputo, B. (2023). Epic-kitchens-100 unsupervised domain adaptation challenge: Mixed sequences prediction. **arXiv:2307.12837**. [Report]

Awards, Fellowships, & Grants

2026 **Outstanding Reviewer**, ECCV 2026

2026 **IscrC “HiERO” grant (68k compute hours)**, CINECA

2026 **2nd place at Ego4D Step Grounding Challenge**, CVPR 2026

2026 **Silver Reviewer**, ICML 2026

2025 **IscrC “HighHAU” grant (80k compute hours)**, CINECA

2025 **Top Reviewer**, NeurIPS 2025 (conference registration fee waiver)

2025 **Doctoral Consortium**, selected as a mentee at ICCV 2025 (conference registration fee waiver + \$1200 travel expenses reimbursement)

2025 **IscrC “EgoProc” grant (56k compute hours)**, CINECA

2023 **2nd place at EPIC-Kitchens UDA Challenge**, CVPR 2023

Teaching Experience

2026 **Modelli per l’AI (Models for AI - Module on image detection and segmentation)**, Teacher

2024-2026 **Advanced Machine Learning**, Teaching Assistant

2024-2026 **Data Analysis and Artificial Intelligence**, Teaching Assistant

2024-2026 **Algorithms and Data Structures (Italian)**, Teaching Assistant

2024-2025 **Machine Learning and Deep Learning**, Teaching Assistant

Co-Supervised M.Sc. Students

2026 **Andrea Zenotto**, Learning the hierarchy of human activities

2025-2026 **Mattia Sabato**, Higher-Order Message Passing for Structured Video Understanding of Human Dynamics

2025 **Federico Castriotta**, Learning the hierarchy of human activities in multi-view scenarios

2023-2024 **Domenico Mereu**, MARE-Graph: Multimodal Action Recognition in Egocentric video with Graph Neural Network

2024-2024 **Giuseppe Atanasio**, HIMU-MAE: Exploiting Head-mounted Inertial Measurement Unit with Masked Autoencoders for Egocentric Vision

2023-2024 **Amirshayan Nasirimajd**, Sequential Domain Generalisation for Egocentric Action Recognition

Outreach And Community Engagement

CONFERENCE ABSTRACTS / SHORT PAPERS

Peirone, S. A., Pistilli, F., & Averta, G., (2025). Exploiting hierarchy and compositionality to understand human behavior. Oral presentation at the *ACVR 2025 workshop*, organized in conjunction with ICCV 2025.

Peirone, S. A., Pistilli, F., & Averta, G., (2025). HiERO: understanding the hierarchy of human behavior enhances reasoning on egocentric videos. Oral presentation at the *SG2RL 2025 workshop*, organized in conjunction with ICCV 2025.

- Peirone, S. A., Pistilli, F., Alliegro, A., Tommasi, T., & Averta, G., (2025). Learning reusable concepts across different video understanding tasks. Poster presented at the *EgoVis* and *Visual Concepts workshops*, organized in conjunction with CVPR 2025.
- Peirone, S. A., Pistilli, F., & Averta, G., (2025). Reasoning on hierarchical representation of human behavior from Ego-videos. Poster presented at the *EgoVis workshops*, organized in conjunction with CVPR 2025.
- Peirone, S. A., Goletto, G., Planamente, M., Bottino, A., Caputo, B., & Averta, G. (2024). Egocentric zone-aware action recognition across environments. Poster presented at the *FOCUS Workshop*, organized in conjunction with ECCV 2024.
- Peirone, S. A., Pistilli, F., Alliegro, A., & Averta, G., (2024). A Backpack Full of Skills: Egocentric Video Understanding with Diverse Task Perspectives. Poster presented at the *EgoVis workshop*, organized in conjunction with CVPR 2024.
- Peirone, S. A., Planamente, M., Caputo, B., & Averta, G. (2023). Test Time Adaptation for Egocentric Vision. Short paper at the *3rd National Conference on Artificial Intelligence (Ital-IA)*.

SUMMER SCHOOLS

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|------------|---|----------------------|
| July 2025 | International Computer Vision Summer School , Computer Vision for Spatial Intelligence | <i>Scicli, Italy</i> |
| Sept. 2023 | ELLIS Summer School , Large-Scale AI for Research and Industry | <i>Modena, Italy</i> |

PEER REVIEW

CVPR, ICCV, ECCV, NeurIPS, ICML, BMVC, IJCV, ICRA, TETC, TMM, TPAMI